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Exploring Jet Substructure in Public LHC Data

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AGRADECIMENTOS

RESUMO

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LIST OF ABBREVIATIONS AND SYMBOLS

1 INTRODUCTION

1.1 Context and motivation

The Large Hadron Collider (LHC) at CERN brings the highest energy proton-proton collisions ever conceived in a laboratory and, with them, an enourmous sample of strongly interacting final states. The majority of these collisions are governed by the Quantum Chromodynamics (QCD), the gauge theory of the strong interaction. The comprehension and study of these collisions are not only a prerequisite for surpressing backgrounds in searches for new phenomena but also the structure of the hadronic final state can be examined and compared to theory directly, at scales where the coupling is weak enough for peturbation theory to be applied and yet the dynamics still remain rich.

The most prominent objects in these final states are jets. Jets are collimated sprays of hadrons that emerge from the fragmentation of high energy quarks and gluons. A jet is not a fundamental object but instead a definition. A clustering algorithm specifies which detector inputs are grouped together and a recombination scheme specifies how momentum is assigned to a result. This defines a jet, from which an interesting and important question then arises. What does the radiation inside the jets look like? This domain is called jet substructure, which over the past decade has grown from a collection of tagging tools into a precision probe of QCD dynamics in its own right.

In order to organise and look at the internal structure of a jet, one can use the Lund jet plane [1]. The Lund jet plane is a two dimensional map of a jet's branching history on which soft, collinear and non perturbative regions of QCD radiation occupy distinct territories. This map records where the jet's internal emissions occur, but it sets aside an intriguing piece of information, the orientation of each splitting relative to the one before. This orientation is not random. Through the linear polarisation of intermediate gluons, successive splittings are correlated in their relative azimuthal angle, producing a characteristic modulation in the distribution of that angle. This is a genuine quantum interference effect which survives only because the intermediate gluon is described by an amplitude rather than a probability, and it is absent from any picture that treats successive emissions as independent classical branchings.

These collinear spin correlations are well understood theoretically and are implemented in modern parton shower algorithms, yet establishing them cleanly in collision data has proved rather difficult. The reason is that the effect carries opposite signs in different splitting channels. The $g \to q\bar{q}$ and $g \to gg$ channels contribute with opposite sign, so a measurement that sums all channels sees the two contributions partially cancel. This interference is not absent from such an inclusive measurement. It is diluted and the apparent size reflects the accidental flavour composition of the sample rather than the strength of the underlying physics.

This is the obstacle the present work addresses. A meaningful measurement of the effect requires knowing which splitting produced each emission. However, the flavour of an individual splitting inside a jet is not directly observable and must be inferred. Working entirely on public CERN Open Data [2], more precisely with CMS Open Data [3], this dissertation builds a reproducible pipeline from the released samples through Lund declustering to the azimuthal spin observable, and introduces a flavour selection that isolates the $g \to q\bar{q}$ channel. The resulting modulation is reported not as a single number but as a function of channel purity achieved by that selection, a presentation that separates the physics of the

interference from the performance of the selection and allows the magnitude of the effect to be read off in the limit of a pure sample.

The Lund declustering angle $\Delta\psi_{12}$ is not the only way to reach this effect. The same $\cos 2\psi$ modulation also imprints itself on the flow of energy within a jet, and can be accessed through a three-point energy correlator, which probes the relative azimuthal orientation of energy deposited by triplets of particles without reference to any clustering history. The two observables are complementary: one reads the effect from the reconstructed branching sequence, the other from the energy flow directly, and each carries a different exposure to the choices and imperfections of the measurement. This work develops both, treating the Lund observable as the primary measurement and the energy correlator as an independent cross-check on the same underlying physics.

The fact that this analysis rests on public data is not incidental. The CERN Open Data Portal [2] has made it possible for work outside the experimental collaborations to produce meaningful results and to carry out genuinely new measurements. The present work is intended as a contribution in that spirit. It builds an end to end pipeline from a single CMS dataset to the final measurements, establishing along the way a workflow for accessing, handling and measuring with CMS Open Data.

1.2 Objectives

The main aim of this dissertation is to measure collinear spin correlations in QCD jets using public LHC data and to establish the flavour selection that such a measurement requires in order to be meaningful. For the purpose of achieving that goal, the work is organised around a sequence of objectives. The first is to build a reproducible pipeline over CERN Open Data samples capable of extracting the necessary variables to begin a jet substructure analysis. This means that the data samples must contain jet constituent information, thus the data format chosen must retain that level of detail, ruling out the more compact samples in favour of one that preserves per-constituent quantities.

With that pipeline in place, the second objective is to construct the primary and secondary Lund jet planes and to verify that they reproduce the known features of QCD radiation, since this agreement is what establishes that the declustering is sound before any finer observable is built on top of it. Following that, the third objective is to implement the azimuthal observable $\Delta\psi_{12}$ between the hardest primary and secondary branchings, and to study its distribution both with and without grooming, recovering the inclusive result before attempting to sharpen it.

The fourth objective is the central one, and the point at which this work departs from a purely inclusive analysis. Because the spin-interference signal carries opposite signs in the $g \to q\bar{q}$ and $g \to gg$ channels, an inclusive measurement dilutes it. Isolating a single channel is therefore not a refinement but a precondition for measuring the effect at its true strength. This objective is thus to develop a flavour selection capable of enriching the $g \to q\bar{q}$ channel, and to quantify both how well it performs and how pure a sample it can reach in principle. The fifth objective builds directly on it. It is to measure $\langle \cos 2\psi_{12} \rangle$ as a function of the channel purity delivered by that selection and to establish the value characteristic of a pure $g \to q\bar{q}$ sample.

Finally, the work aims to assess the systematic uncertainties affecting this measurement and to identify the dominant limitations of the approach, so that the extensions it opens up can be set out on solid ground.

1.3 Thesis outline

Chapter 2 develops the theoretical material on which the measurement rests, such as the elements of QCD relevant to jet formation, the definition and reconstruction of jets and the algorithms used to build and decluster them, the Lund jet plane, and the origin of collinear spin correlations together with the two observables through which this work accesses them. Chapter 3 turns to the data and the analysis, and covers the CERN Open Data resources and the CMS detector, the data format and dataset used, and the pipeline that runs from the released samples through Lund declustering to the azimuthal observable, the flavour selection, and the treatment of systematic uncertainties. Chapter 4 presents the validation of the pipeline and the measurement itself, and discusses the results in the light of the expectations set out in Chapter 2. Finally, Chapter 5 returns to the objectives of Section 1.2, states what has been established and with what confidence, and outlines the extensions that the present framework makes possible.

2 THEORETICAL BACKGROUND

2.1 Quantum Chromodynamics

Quantum Chromodynamics (QCD) is the quantum field theory of the strong interaction, a non-Abelian gauge theory built on the colour symmetry group $SU(3)$ [4]. Its fundamental degrees of freedom are quarks, which carry colour charge, and gluons, the gauge bosons that mediate the interaction. The non-Abelian character of the theory has a decisive consequence absent from Quantum Electrodynamics. The gluons themselves carry colour, and therefore couple directly to one another. To a large extent, the dynamics of the strong interaction are the dynamics of this gluon self-coupling, and the two properties of QCD that govern everything in this chapter both follow from it.

2.1.1 Asymptotic freedom and confinement

The first property is asymptotic freedom. The strong coupling α_s decreases as the momentum transfer of an interaction increases, so that at the high scales probed in LHC collisions the quarks and gluons interact weakly and perturbation theory applies [5, 6]. The running of the coupling is captured at leading order by

$$\alpha_s(Q^2) = \frac{\alpha_s(\mu^2)}{1 + \alpha_s(\mu^2)\,\frac{33 - 2n_f}{12\pi}\,\ln\frac{Q^2}{\mu^2}}, \tag{2.1}$$

where Q is the scale of the process, μ a reference scale, and n_f the number of active quark flavours. The positive sign of the coefficient $33 - 2n_f$, which holds for any physical number of flavours, is what makes the coupling fall at large Q. It is a direct imprint of the gluon self-interaction, and its discovery established that a perturbative treatment of hard processes is justified. It is convenient to trade the reference scale for a single dimensionful constant Λ_{QCD}, the scale at which the perturbative coupling formally diverges, of an order of a few hundred MeV. In terms of it the coupling is small for $Q \gg \Lambda_{\text{QCD}}$ and grows strong as Q approaches it.

This growth is the second property called confinement. At low momentum transfer α_s becomes large and colour-charged particles cannot be isolated. A quark or gluon produced in a hard scattering does not propagate freely to a detector, because as it separates from other colour charges, the energy stored in the colour field between them grows with distance until it becomes energetically favourable to create a new quark-antiquark pairs from the vacuum. The original parton is thereby dressed into a spray of colour-neutral hadrons. This is why the strongly interacting final state reaches the detector not as partons but as the collimated sprays introduced in Chapter 1, and why any comparison between a perturbative calculation and a measurement must bridge the divide between the partonic and hadronic descriptions.

2.1.2 Soft and collinear radiation

The bridge between the hard parton and the observed hadrons is built from successive emissions, and the pattern of those emissions is far from uniform. When a parton radiates a gluon, the emission probability

at first order in the coupling takes the schematic form

$$d\mathcal{P} \propto \frac{2\alpha_s C}{\pi} \frac{d\theta}{\theta} \frac{dE}{E}, \tag{2.2}$$

where E and θ are the energy and emission angle of the radiated gluon and C is a colour factor. The two factors carry the characteristic structure of QCD radiation. The factor $d\theta/\theta$ diverges as $\theta \to 0$ meaning that emissions are enhanced when the gluon is radiated close to the direction of its emmiter. This is known as the collinear divergence. The factor dE/E diverges as $E \to 0$ meaning that emissions are enhanced when the gluon is soft. This is known as the soft divergence. Together they mean that a hard parton radiates predominantly softly and at small angles, and that number of such emissions is, in a precise logarithmic sense, large.

The collinear structure is described more completely by the Dokshitzer–Gribov–Lipatov–Altarelli–Parisi (DGLAP) splitting functions [4], which give the probability that a parton of one type splits into two, as a function of the momentum fraction z carried by one of the products. For the three splittings relevant to this work they read, at leading order,

$$P_{q\to qg}(z) = C_F \frac{1+z^2}{1-z}, \tag{2.3}$$

$$P_{g\to gg}(z) = C_A \left[\frac{z}{1-z} + \frac{1-z}{z} + z(1-z)\right], \tag{2.4}$$

$$P_{g\to q\bar{q}}(z) = T_R \left[z^2 + (1-z)^2\right], \tag{2.5}$$

with colour factors $C_F = 4/3$, $C_A = 3$ and $T_R = 1/2$. The distinction between the gluon splitting channels of Eqs. (2.4) and (2.5), one producing a pair of gluons and the other a quark-antiquark pair, will prove central. The two channels differ not only in these rates but in the spin structure of their emissions, and it is that difference which the spin observables of Section 2.4 are designed to probe.

These divergences are not a pathology. They are integrable, and in any physical, infrared-safe observable the divergent real emissions cancel against corresponding virtual corrections, leaving a finite result. This cancellation is the requirement that shapes the very definition of a jet in Section 2.2. Their logarithmic structure, however, is physical and measurable, and it is exactly the structure that the Lund jet plane of Section 2.3 is built to expose. By mapping each emission onto logarithmic axes of angle and transverse momentum, it turns the double logarithmic enhancement of equation (2.2) into an approximately uniform density, against which finer correlations stand out.

2.1.3 Parton showers and hadronization

Because a single hard parton typically radiates many times before hadronizing, the emissions can be organised into a sequence orderred from the hardest, widest-angle splittings down to the soft and collinear ones. This sequence is what is known as a parton shower. Modern Monte Carlo event generators implement this sequence probabilistically through the splitting functions of equations (2.3)-(2.5), evolving each parton down from the hard scale of the collision to a low scale of order 1 GeV, at which the perturbative description breaks down and confinement takes over. Below that scale the partons are converted into hadrons by phenomenological hadronization models, tuned to data rather than derived from first principles.

This picture, a hard, perturbatively calculable core, dressed by a shower of increasingly soft and collinear emissions and finally hadronized, underlies the whole of the analysis that follows. The observables studied in this work are built to read the perturbative shower through the hadrons that reach the detector, and for this reason the measurement is deliberately confined to a hard kinematic regime, where the perturbative component dominates and non-perturbative corrections are comparatively suppressed.

2.2 Jets

As settled in Section 2.1, a coloured parton produced in a hard scattering does not reach the detector directly because it radiates and hadronizes before arriving as a collimated spray of hadrons. A jet is then an object, which by grouping those hadrons and combining their momentum back together inverts the process approximately and estimates the momentum of the orininal parton that produced them. Therefore, a jet is a reconstructed physical object, built from a construction algorithm, with properties that depend on this specific algorithm used to build it. This section sets out that the requirement which any jet definition must satisfy, in order to be usable in peturbative QCD, and three algorithms, used in this work and built on that requirement.

2.2.1 Jet algorithms and infrared–collinear safety

A jet is built using two components. A jet algorithm establishes which inputs, from detector-level energy deposits, reconstructed particles, or, in a calculation, partons, are grouped together into a jet. A recombination scheme picks up those grouped inputs and specifies how their momenta is combined in order to compute the momentum of the jet. Across this work, the inputs are the particle-flow candidates, described in Chapter 3, and momenta are combined in the E-scheme, in which four-momenta are simply added. This scheme is Lorentz invariant and returns jets with non-zero mass.

Though, every conceivable grouping prescription is not admissible. The requirement necessary and the one that eliminates most of them is infrared and collinear (IRC) safety. In order for an observable to be infrared safe is that its value most remain unchanged by the addition of an arbitrarily soft emission. For it to be collinear safe, this value most also reamin unchanged if one of its inputs is replaced by two collinear inputs sharing its momentum. So, a jet algorithm is IRC safe when the set of jets it returns is stable under both of these operations. A soft particle must not create or destroy a jet and a collinear splitting must not change how particles are assigned to jets.

This requirement is not optional because of what follows directly from Section 2.1. The emission probability of equation (2.2) diverges in both the soft and the collinear limits. In any IRC-safe observable these divergent real emissions cancel against equally divergent virtual corrections, leaving a finite result. Opposingly, in an IRC-unsafe observable, that cancellation is ruined and a fixed-order peturbative prediction returns infinity rather than an finite number. An IRC-unsafe jet definition is therefore not merely inconvenient but it also cannot be compared with perturbative QCD at all. Since the purpose of measuring jets is to confront the theory with data, IRC safety is a precondition and not a refinement.

In the historical development of jet algorithms this criteria has been conclusive. The first definitions that appeared were cone algorithms, which grouped particles falling within a fixed angular radius of a candidate axis. These algorithms were simple and seed-based and were not IRC safe because a soft particle could seed a new candidate cone, or a collinear splitting could change which particles acted as

seeds, thus either these flaws could alter the resulting jets. A lot of effort was put in order to fix them and eventually producing seedless variants that restored collinear safety. Alternatively, another family of algorithms, the sequential recombination algorithms, are IRC safe by construction and this work utilised them.

Since the same mechanism reappears when flavour is assigned to jets in Section 2.2.4, it is worh making concrete how such a problem arises. Consider a seed-based cone algorithm applied to an event with two hard particles whose separation is slightly greater than the cone radius, so that each seeds its own jet. By adding a single soft particle between them, it can seed a third cone that overlaps both, merging the two hard particles into one jet where before there were only two, which, again, it is an infrared-unsafe change, since an arbitrarily soft emission has altered the hard jets. A collinear splitting produces the collinear-unsafe failure. Replacing one hard seed by two collinear particles, each carrying half its transverse momentum, may leave neither above the seed threshold, so that a jet which existed before the splitting is absent after it. Therefore, in both cases, the divergent emissions of equation (2.2) feed directly into the jets themselves, and the perturbative prediction diverges with them.

Going back to the family introduced before, the sequential algorithms avoid both failures by never relying on seeds or on fixed geometric cones. In these algorithms, jets are built by repeatedly merging inputs according to a distance measure, so that soft or collinear additions change only the order in which already nearby inputs are combined and never the final assigment of hard momentum jets. This structural difference is the reason why this entire family is IRC safe and it is why it is recommended for any measurement to be compared with theory over the cone algorithms.

A sequential recombination algorithm is defined by a distance measure between each pair of inputs and between each input and the beam. The generalised-k_T family [7] writes these as

$$d_{ij} = \min\left(p_{\mathsf{T},i}^{2p},\, p_{\mathsf{T},j}^{2p}\right) \frac{\Delta R_{ij}^2}{R^2}, \tag{2.6}$$

$$d_{iB} = p_{\mathsf{T},i}^{2p}, \tag{2.7}$$

where $\Delta R_{ij}^2 = (y_i - y_j)^2 + (\phi_i - \phi_j)^2$ is the squared angular separation of the inputs in rapidity y and azimuth ϕ, R is the jet radius parameter, and the integer p selects the algorithm. The algorithm, at each step, finds the smallest of all the distances. f it is a d_{ij}, the inputs i and j are recombined into a single input. If it is a d_{iB}, the input i is declared a final jet and removed. This procedure repeats until no inputs remain. The returned jets are IRC safe for any integer value of p, but the value of p changes extremely how the clustering proceeds. The three choices $p = -1$, $p = 0$ and $p = +1$ give, respectively, the anti-k_T, Cambridge/Aachen and k_T algorithms, of which the first two are the tools this work relies on and are described in the following subsections.

In all three of those algorithms, the parameter R appears and sets the angular reach of the clustering through the ratio $\Delta R_{ij}^2/R^2$ in equation (2.6). A smaller R yields narrower jets that resolve nearby structure but also capture less of a single parton radiation. A larger R collects more of that radiation at the price of admitting more contamination from unrelated soft activity. The most appropriate choice depends on what we intend to use the jets for, and the two algorithms this work relies on are used with different values of R for correspondingly different goals. This point is, again, taken up in Sections 2.2.2 and 2.2.3.

2.2.2 The anti-k_T algorithm

The choice $p = -1$ in Eqs. (2.6) and (2.7) defines the anti-k_T algorithm [8]. With this value the inter-particle distance becomes

$$d_{ij} = \min\left(\frac{1}{p_{\mathsf{T},i}^2}, \frac{1}{p_{\mathsf{T},j}^2}\right)\frac{\Delta R_{ij}^2}{R^2}, \tag{2.8}$$

so that the distance scales with the inverse squared transverse momenta of the inputs. This makes the smallest distances, and thus the first recombinations, commit to the hardest particles. The clustering is therefore driven by the hard inputs. So, a hard particle accumulates the softer inputs around it before soft inputs cluster among themselves.

The name ”anti” of this algorithm reflects this inversion. In the k_T algorithm, the one with the choice of $p = 1$, the distance scales with the positive powers of transverse momentum, making the soft unputs cluster first and the jet being assembled from the bottom, following the inverse of the physical shower order. The anti-k_T algorithm does the reverse of it. When taking $p = -1$, the soft-first ordering of k_T turns into a hard-first one. Effectively, it is the k_T algorithm with the role of soft and hard interchanged, and the ”anti” part shows exactly that inversion. The two sit at opposite ends of the generalised family, with Cambridge/Aachen, free of any transverse momentum dependence, between them.

There is a geometric consequence because of this ordering. A hard particle with no other hard particle within an an angular distance $2R$ gathers all softer inputs within a distance R of it, producing a jet that is exactly circular in the (y, ϕ) plane. When two hard particles are closer than $2R$, the border between their jets is set by their relative transverse momenta with the harder one retains the larger share, but the jets remain regular in shape. Due to soft radiation being last in the clustering, it cannot deform these borders. They are spread across the hard jets without chaging their shape.

The region of the (y, ϕ) plane from which a jet gathers its inputs is a useful way to describe this stability. This area can be probed operationally by adding a large number of infinitesimal soft particles to an event and record which jet each of these particles is assigned. For the majoroty of the algorithms, the resulting areas are irregular and fluctuate from jet to jet, because soft particles impact the clustering. However, for the anti-k_T algorithm, where soft particles are clustered last and cannot reshape the hard jets, the areas are close to the ideal value πR^2 and fluctuate little. A stable and expected area is what makes the subtraction of pile-up and underlying-event contamination tractable, since that contamination scales with the area over which each jet collects it.

This regularity is what made the anti-k_T algorithm the standard jet definition at the LHC. Since the jet border is stable against soft radiation, the area over which a jet collects contamination from pile-up and the underlying event is well defined and almost constant, and the corresponding energy corrections are simple to calibrate. Therefore, the anti-k_T algorithm combines the IRC safety guaranteed for the whole generalised-k_T family with a jet form that is simple to treat experimentally. This combination thus provides regular borders, which the other two generalised-k_T algorithms do not.

The importance of the anti-k_T is also a matter of history. When the LHC started taking data, an IRC safe jet definitions was needed, so that measurements could be compared with the increasingly precise perturbative calculations that became available. The algorithm was also required to be fast enough to run along the very large event samples which the LHC would produce. The anti-k_T algorithm, together with its efficient implementation in the FastJet package [9], met both needs, and it was adopted as the

baseline jet definition by the LHC experiments, a role it has held since.

It is because of this that the anti-k_T algorithm was used in this work to define the jets, with radius parameter $R = 0.8$. Jets constructed in this way are conventionally denoted as AK8. Howver, the algorithm does not provide a physically meaningful account of how the jet was assembled. This is becacause its sequence is ordered by hardness rather than by angle and the intermediate recombinations bear no correspondence to the sequence of physical splittings in a parton shower, which is ordered in angle. The clustering history of an anti-k_T jet is, in this sense, an artefact of the algorithm rather than a record of the underlying radiation. Recovering that history, which is what the substructure observables of this work require, calls for a different member of the family, applied to the constituents of the jet once it has been defined. That algorithm is the Cambridge/Aachen algorithm and it is the subject of the next subsection.

2.2.3 The Cambridge/Aachen algorithm

From equation 2.6 and equation 2.7, if one chooses $p = 0$, the Cambridge/Aachen (C/A) algorithm is defined [10]-[11]. With this choice, the transverse momentum factors drop out of the distance measure entirely, leaving the equation to a purely angular form:

$$d_{ij} = \frac{\Delta R_{ij}^2}{R^2}. \tag{2.9}$$

The clustering process with this algorithm proceeds by repeatedly merging the pair of inputs closest in angle, independently of how hard or soft they are. Where the anti-k_T algorithm is driven by the hardest inputs, the C/A algorithm orders them by angular separation alone. This means that the first merges are the most collinear and the last the widest.

This type of angluar ordering is what makes the C/A algorithm the optimal tool choice for evaluating jet substructure. As discussed in Section 2.1, the parton shower is ordered in angle, with the widest angle splittings happening earliest in the physical branching sequence. Therefore, the C/A algorithm mirrors, in reverse, the order in which the shower developed. By undoing the clustering process one step at a time, an approximation of the branching sequence is retraced from the widest splitting to the narrowest splitting, and each step exposes a single emission, characterised by the angular separation ΔR and the relative transverse momentum k_t of the two branches it separates. This agreement between the C/A history and the physical shower is what the Lund jet plane, discussed in Section 2.3, is built to exploit, and it is the reason jet substructure observables are extracted from the C/A history rather than the anti-k_T one.

The declustering process can be asserted explicitly. Starting from the single C/A object that contains the whole jet, one undoes the last recombination, splitting it into two subjets a and b, labelled so that $p_{t,a} > p_{t,b}$, meaning a is the harder branch and b the softer branch. This softer branch is taken as the emission and then, the declustering process, follows the harder branch a, repeating the previous step of splitting it. Iterating along the hard branch until it can no longer be declustered traces out the primary sequence of the jet. This is a ordered list of emissions, from the widest and earliest to the most collinear and latest.

Every splitting in this sequence is characterised by a small set of kinematic variables. The angular separation of the two branches ΔR_{ab}, and their relative transverse momentum $k_t = p_{t,b}\,\Delta R_{ab}$, together locate the emission on the two scales that matter. This characterises the splitting by how hard and how hard it is. A third variable, the momentum fraction $z = p_{t,b}/(p_{t,a} + p_{t,b})$, computes how the momentum

was distributed between the emission and the emitter. These are the quantities from which the Lund jet plane of section 2.3 is constructed, and computing them at every declustering step is the immediate goal of the C/A history.

By following only the harder branch, at each step, one captures the primary emissions, although it discards the structure within the softer branches. In order to recover that structure, it is necessary to decluster it recursively. By choosing a primary emission, the softer branch b is itself declustered in the same way, exposing the emission within it and generating a secondary sequence of emissions associated with that primary splitting. The primary sequence records how the leading parton radiated and the secondary sequences record how those radiated partons radiated in turn. In this work, both primary and secondary emissions' observables are needed in order to compare the orientation of a primary splitting with that of the secondary splitting it seeds.

In practice, the anti-k_T and C/A algorithms are applied in succession and complement each other. A jet is first defined with the anti-k_T algorithm, since it fixes its constituents and its regular boundary. Those constituents are then extracted and reclustered with the C/A algorithm, in order to obtain an angularly ordered history that can expose the emissions within a jet. In this work, the radius parameter of the C/A reclustering step was set to $R = 1$, larger than the anti-k_T value of $R = 0.8$, to ensure that all the constituents of the original jet are gathered into a single C/A object and that the declustering history is complete with no emission lost at the edge. Thankfully, the kinematics of the jet are unchanged by performing this operation because the same constituents with the same momentum are involved, however they now become organised into a sequence from which the physical splittings can be read.

The division of labour between the two algorithms is thus deliberate. The anti-k_T algorithm supplies a jet with a stable, regular boundary, well suited to the energy calibration and pile-up subtraction discussed in Section 2.2.2. The C/A algorithm applied to that jet's constituents supplies a history that mirrors the physical showering sequence with angularly ordered branching history which substructure requires. Neither algorithm alone would suffice. The anti-k_T algorithm gives a clean jet but an unphysical history, while the C/A algorithm gives a physical history but an irregular, contamination-prone boundary. Using each where it is strong is what makes the substructure analysis of this work both experimentally robust and physically interpretable.

2.2.4 Flavoured jets and the IFN algorithm

In the previous subsections, the algorithms cluster inputs on the basis of their kinematics alone, which is sufficient for most purposes. However, for the core measurement of this work, kinematics alone are not enough. As foreseen in Chapter 1, the goal is to explore the spin correlations of gluon splitting, and that requires distinguishing the channels of equations (2.4), (2.5) and (2.3), which produce, respectively, a pair of gluons, a quark or anti quark with a gluon and a quark-antiquark pair. The channels differ not only in their rates but also their spin interference signal carries different signs and magnitudes. Evaluating this effect at its real strengh thus require separating channels, and, in addition, that requires flavour assigment to the splittings inside a jet. This raises the obstacle of how can one assign flavour to a jet, or even to a branch inside one, in a way that remains theoretically well defined and experimentally robust.

The most obvious procedure is to sum the flavour of a jet's constituents, by counting a quark as $+1$ and an antiquark as -1 for each quark flavour separatelly, so that the net flavour of a jet is that of the parton which initiated it. Although this procedure is simple, it is not IRC safe and the root of the problem is precisely the $g \to q\bar{q}$ splitting. A gluon carries no net flavour, but it can split into a quark-antiquark pair that is soft or

wide in angle or even both. This quark and antiquark can then be assigned to different jets, or even to different branches of the same jet, making each acquire a net flavour that the gluon never possessed. Because the splitting can be arbitrarily soft, the missassigment occurs with a probability that diverges in the same manner as the emission spectrum of equation (2.2), and no virtual correction is assigned to cancel it. This is the same mechanism for kinematic observables, identified in Section 2.2.1 but now for the flavour. Here, this simple procedure is ruined by exactly soft and collinear radiation that QCD produces most abundantly, and a fixed order calculation of it fails to converge. Thus, flavour assigned sloppily is not a physical quantity and it cannot be compared with theory.

Restoring IRC safety to jet flavour has been pursued for two decades. The first safe prescriptions achieved IRC safety by making the clustering distance itself depend on the flavour of the objects being combined [12], which succeeded in supressing the divergence, but they costed the produced jets to not have the kinematics of the fundamental algorithm. This means that a flavour aware jet and its equivalent anti-k_{T} or C/A jet were no longer the same object. For a measurement that requires both kinematics of a jet and the flavour of its structure, that cost is too high. In this work, the approach adopted was using the Interleaved Flavour Neutralisation (IFN) [13], which was designed exactly to avoid that issue. IFN leaves the kinematics of the clusturing entirely untouched and treats flavour as a separate layer that runs alongside the normal clustering sequence. The jets, which are returned by this algorithm, are the identical to the C/A jets of Section 2.2.3. This includes constituents and momentums. What IFN gives in addition is a flavour label for each, computed by a procedure that is stable against soft and collinear emissions.

The principle behind this new procedure is neutralisation. While the clustering occurs, IFN tracks down the flavour carried by every object in the sequence, and, at each stage, asks wether a flavoured object can be matched to a nearby object of opposite flavour, which is the partner who would have originated the common $g \to q\bar{q}$ splitting. Whenever it finds such a pair, the two flavours are neutralised against each other, so that the ingenuine flavour is cancelled at the moment it would otherwise be passed up the clustering tree and inherited by a jet. This makes the quark and antiquark born of the same soft or collinear gluon splitting reunite in flavour, even though their kinematics remain whatever the C/A algorithm placed them. The matching is ruled by a flavour distance that weighs how soft and how collinear a candidate pair is, so that it is exactly the soft and collinear configurations that are neutralised, while genuinely distinct hard flavours are left in place.

What IFN brings to this work is that it does not merely give a prescription that labels the final jet. It exposes the flavour at every step of the clustering and not only at the end. This is the feature that the present analysis required. he clustering of a C/A history produces at each step a splitting of one object into two branches. IFN attaches to those branches a flavour label that is IRC safe and, therefore, meaningful at fixed order in perturbation theory. The abstract distinction between the discussed channels in Eqs. (2.4), (2.5) and (2.3) is thereby made concrete, and it can be read off at each splitting in the history of a jet. It is on this per splitting flavour that the channel selection at the heart of the measurement is built. The algorithm is distributed as a plugin to the FastJet package [9], and the configuration adopted here, together with the way its per-splitting labels are combined into the selection itself, is set out in Chapter 3.

2.3 The Lund jet plane

Described in Section 2.2.3, the declustering of the C/A history, links with every jet an ordered sequence of emissions, each carrying an angular separation ΔR, a relative transverse momentum k_t and a momentum fraction z. This sequence, on its own, is just a list of numbers. Jets have different numbers of emissions

and the raw variables reach many orders of magnitude. However, the Lund jet plane [1] can represent this list into something clear, when mapping each emission to a single point in a two-dimensional plane whose coordinates are chosen so that the structure of QCD radiation is exposed in its simple form.

2.3.1 Coordinates and the uniform density

The procedure to construct the Lund jet plane is direct. For each jet, the primary declustering of the C/A history, of Section 2.2.3, yields a list of emissions. Every emission in that list is mapped to a single point on the plane using its angular separation ΔR and relative transverse momentum k_t as coordinates. When repeating this for every emission in a sample and overlaying the resulting points, it produces a two-dimensional distribution, which is then normalised to the number of jets inside the sample, giving the average density of emission per unit area. It is this averaged distribution, rather than the pattern of any single jet, that is referred to as the Lund jet plane and that carries the statistical information exploited below.

Each primary emission is placed on the plane at the coordinates:

$$\left(\ln \frac{1}{\Delta R},\, \ln k_{\mathsf{T}}\right), \tag{2.10}$$

the horizontal axis measures the inverse angle of the splitting and the vertical axis its hardness, both logarithmically. The choice of logarithmic axes is not cosmetic. Going back to Section **??**sec:qcd) that the emission probability of a gluon emission carries the double structure $d\mathcal{P} \propto \alpha_s\, d\theta/\theta\; dE/E$, which is one factor for the collinear enhancement and one for the soft enhancement. When written in terms of Lund coordinates of equation (2.10), each of these factors becomes flat. The measure $d\theta/\theta$ is simply $d\ln(1/\Delta R)$, and the energy factor maps onto $d\ln k_{\mathsf{T}}$. The doubly divergent emission spectrum of the raw variables therefore becomes, on the Lund plane, an approximately uniform density of emissions per unit area,

$$\rho \;=\; \frac{1}{N_{\text{jet}}}\,\frac{dn_{\text{emissions}}}{d\ln(1/\Delta R)\; d\ln k_{\mathsf{T}}} \;\approx\; \text{const} \times \frac{\alpha_s}{\pi}, \tag{2.11}$$

up to the slow running of α_s and the effects discussed below. It is this flatness that makes the plane so useful against a uniform background. Any physical effect beyond the leading soft-collinear approximation therefore appears as a localised enhancement or suppression, rather than being obscured by the rapidly falling raw spectrum. This uniform density is precisely what is observed in practice. Figure 2.1 shows the average primary Lund plane obtained from a simulated sample of QCD jets. Through the perturbative majority of the plane the density is approximately constant, as anticipated by equation (2.11), falling away only towards the boundaries of the kinematically accessible region.

2.3.2 Regions of the plane

The reachable area of the plane is delimited, and its interior can be divided into different regions with distinct physical source, sketched in Figure 2.2. The upper boundary is set by the kinematics of the jet. Every emission cannot be harder than the jet which contains it, so it caps $\ln k_{\mathsf{T}}$ from above. Along the right of the plane, at large $\ln(1/\Delta R)$, lie the collinear emissions, those radiated at small angle to their

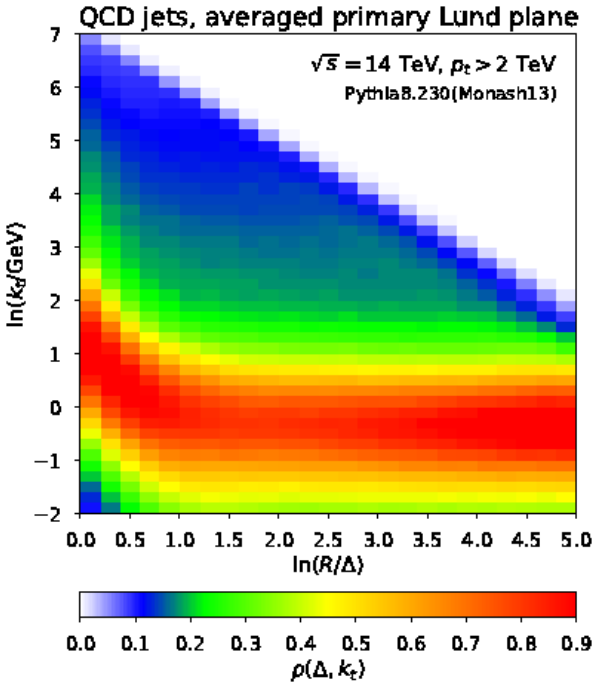


Figure 2.1 – The average primary Lund plane density, ρ, for jets clustered with the C/A algorithm and $R = 1$ having $p_t > 2$ TeV and $|y| < 2.5$, in a simulated QCD dijet sample [1].

emitter. Towards the bottom, at small ln k_T, lie the soft emissions. The perturbative description of Section 2.1 holds across the majority of this region, where k_T is large enough for α_s to remain small, making the density there be the near-constant value of equation (2.11).

The exceptional region in the lower-right corner, at both small angle and low transverse momentum. In that region, k_T falls to the scale of a few hundred MeV, α_s grows large, and the perturbative description breaks down. This region is known as the non-perturbative region and it is populated by hadronisation and by the soft, wide-angle contamination present inside the event. This way, one strong feature of the Lund jet plane is that these non-perturbative effects are confined to a restricted corner of the plane, clearly separated from the perturbative part and not mixed with the whole observable. A measurement can therefofore restrict, by eye or by an specific cut, to the region where theory is applicable.

2.3.3 Grooming and the primary and secondary planes

This clear separation motivates grooming methods. Grooming is the intentional removal of soft, wide-angle emissions that dominate non-perturbative regions of the Lund jet plane. The grooming method used in this work is Soft Drop [14], which walks through the declustering and, at each step, keeps the harder branch unless the softer one satisfies the following:

$$z > z_\mathrm{cut} \left(\frac{\Delta R}{R} \right)^{\beta}, \tag{2.12}$$

in which case both branches are kept and the declustering precedure continues. The parameter z_cut specifies how soft an emission must be to be discarded, and β manages the angular reach of the grooming. The choice $\beta = 0$, adopted here for the Soft Drop in this work, makes the condition purely one of momentum fraction , $z > z_\mathrm{cut}$, independent of angle. In this condition, Soft Drop becomes a modified

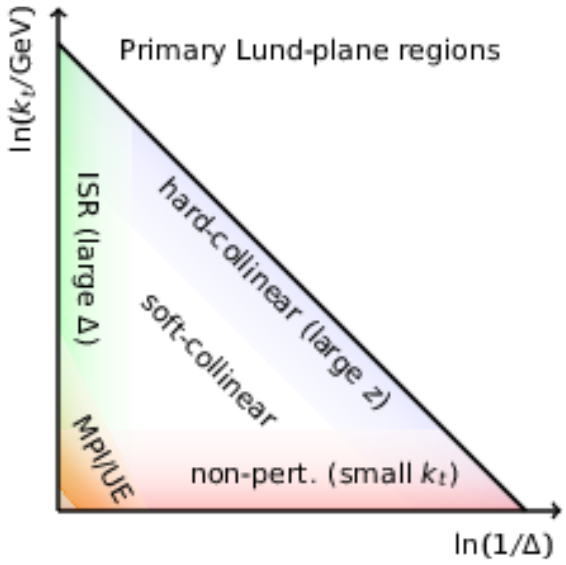


Figure 2.2 – Schematic representation of the di□erent regions of the Lund plane [1].

Mass-Drop Tagger [15]. Applying this grooming method becomes very visible on the Lund jet plane because it removes the emissions in the soft, low k_{T} band, clearing the non-perturbative corner and leaving a plane populated only by the harder splittings.

In the present work, two versions of the Lund jet plane are built and used as follows. The primary Lund plane is built with the emissions along the hard branch of the jet and describes how the leading parton radiated. The secondary Lund plane is built by taking the chosen primary emission and declustering its softer branch in turn, using the same method as the primary Lund plane, as described in Section 2.2.3. The resulting points describe how that radiated parton itself went on to radiate. The two planes together deliver complementary information and both are needed for the spin observables of Section 2.4, which compare the orientation of a primary splitting with that of the secondary splitting it seeds. Figure 2.3 illustrates these two levels of structure for two example jets. The middle row shows the full Lund diagram, in which every emission, primary and secondary, is represented, while the bottom row shows the primary Lund plane alone, retaining a single point for each primary splitting.

2.4 Collinear spin correlations

In the previous sections, the tools assembled describe a parton shower as a sequence of independent splittings, each blind to those before it. This description is incomplete since intermediate parton in a shower carry spin and their spin is correlated with the kinematics of the splitting that produces them. Furthermore, when that parton branches, that correlation is transmitted to the azimuthal orientation of its own decay products. So, successive splittings are therefore not independent, meaning that the plane of one branching is correlated with the plane of the next. These are the collinear spin correlations which this work presents, and they are the subject of this section.

This correlation effect is a quantum interference phenomenon, with no counterpart in a classical picture of sequential radiation, and it shows in a distinctive and robust way. This phenomenon manifests as a modulation of the azimuthal angle between successive splitting planes with a π periodicity instead of 2π, varying directly as a function of $\cos 2\psi$. This section develops that statement in four steps. Section 2.4.1

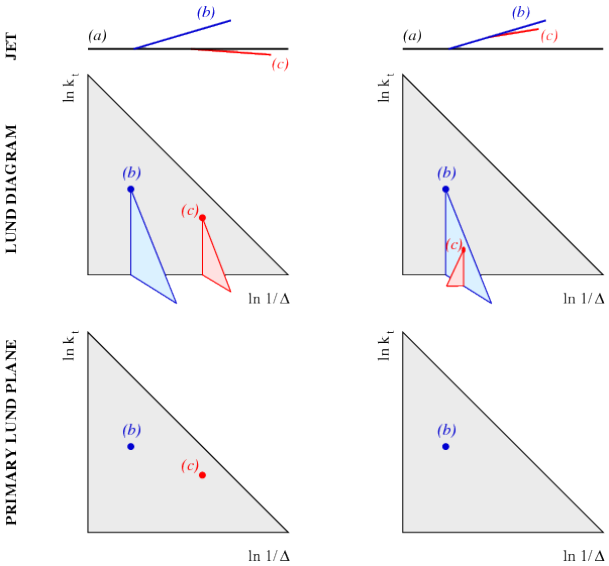


Figure 2.3 – Different representations for two jets. Top: the particles inside the jet. Middle: the full Lund diagram. Bottom: the primary Lund plane [1].

shows the physical mechanism and the origin of the $\cos 2\psi$ modulation. Section 2.4.2 shows how the size and sign of the modulation depend on the splitting channel, the property that makes flavour identification essential to the measurement. Sections 2.4.3 and 2.4.4 then introduce the two observables through which this work accesses the effect. These observables being the Lund declustering angle $\Delta\psi_{12}$, computed from the branching history of Section 2.3, and the three-point energy correlator, which reaches the same physics through the flow of energy within the jet.

2.4.1 Mechanism and the $\cos 2\psi$ modulation

The origin of the correlation lies in the spin of the intermediate gluon [16]. Consider a gluon produced in a first splitting that subsequently undergoes a second splitting of its own. This corresponds to the setup of the primary and secondary Lund planes discussed in Section 2.3. Since the gluon is a spin-one particle, a hard, nearly on-shell gluon possesses only two physical polarisation states, both transverse to its direction of motion. In the collinear limit, these states are most naturally described not as circular polarisations, but as two orthogonal linear polarisations, each representing an oscillation of the field along a fixed axis in the plane transverse to the gluon momentum.

The polarisation of the intermediate gluon is not random. The first splitting, which produced it, selects a preferred transverse axis. Therefore, the gluon emerges linearly polarized with its polarisation correlated with the plane of that first branching. This is the main and essential point. A gluon radiated in a splitting carries a memory of the orientation of that splitting, engraved in the direction of its linear polarisation.

Afterward, that memory is read out by the second splitting. The probability that a linearly polarised glluon branches into a given final state depends on the azimuthal orientation of the second splitting plane in respect with the polarisation axis. Writing ψ for the angle between the first and the second splitting planes, the differential rate acquires a dependence on ψ through this alignment, and the two splittings become correlated in their relative azimuth.

The period of that correlation is fixed by the nature of the polarisation. A linear polarisation is specified by an axis, not by a direction along it. Rotating the polarisation by π returns the same physical state,

since an oscillation along an axis is unchanged by reversing its sense. The azimuthal dependence must therefore be invariant under $\psi \to \psi + \pi$, which forbids a $\cos\psi$ term and admits $\cos 2\psi$ as its leading harmonic. This is the distinctive signature of the effect. The result has to have a modulation of period π rather than 2π, reflecting the headless, axial character of linear polarisation. Collinear spin correlations announce themselves not as any azimuthal dependence, but specifically as a $\cos 2\psi$ dependence.

At second order in the strong coupling and in the collinear limit, the cross section differential in ψ therefore takes the form

$$\frac{d\sigma}{d\psi} \propto a_0 \left(1 + \frac{a_2}{a_0} \cos 2\psi\right), \tag{2.13}$$

where a_0 is the spin-averaged rate, or, in other words, the result one would obtain by ignoring the polarisation of the intermediate gluon, and a_2 is the coefficient of the interference term [16]. The ratio a_2/a_0 measures the strength of the correlation where it vanishes if the intermediate gluon carries no polarisation memory, and its magnitude sets the depth of the $\cos 2\psi$ modulation. Its sign determines the phase, wether the second splitting preferentially aligns with the first splitting plane, giving maxima at $\psi = 0, \pi$, or preferentially lies perpendicular to it, giving maxima at $\psi = \pm\pi/2$. The measurement of these spin correlations reduces, in essence, to the measurement of this single ratio, and the whole of the analysis that follows is directed towards extracting it cleanly.

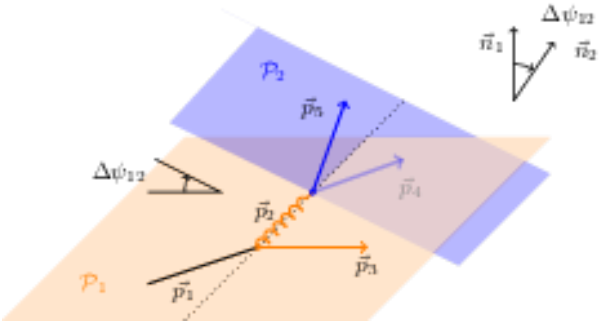


Figure 2.4 – Azimuthal angles are defined between successive splitting planes for the $1 \to 23$ splitting, $\mathcal{P}_1 \supset \{\vec{p}_2, \vec{p}_3\}$, and the $2 \to 45$ splitting, $\mathcal{P}_2 \supset \{\vec{p}_4, \vec{p}_5\}$. The figure also depicts the vectors normal to the two planes, $\vec{n}_1$ and $\vec{n}_2$ [16].

2.4.2 Channel dependence of a_2/a_0

The process described in Section 2.4.1 was made for a generic intermediate gluon, however, the strength and sign of the resulting modulation are not universal. These parameters depend on the nature of the second splitting, or, in other words, on the channel through which the polarised gluon decays. The two channels available are those of equations (2.4) and (2.5). The gluon can spit into a pair of gluons, $g \to gg$, or into a quark–antiquark pair, $g \to q\bar{q}$. Each of these channels read out the polarisation of the parent gluon, but each one of them do so with opposite sign and different strengh [16].

The reason behind this difference comes from the spin of the produced particles. The angular The angular distribution of a splitting depends on how the daughters' spins couple to the polarisation axis of the parent, and this coupling differs between a final state of two vector gluons and one of two spin-half fermions [16]. The difference is that the interference coefficient a_2/a_0 of equation (2.13) appears with one sign for the $g \to gg$ channel and the opposite sign for the $g \to q\bar{q}$ channel, where one channel prefers the alignment of the two splitting planes, the other a perpendicular configuration.

This sign difference is the main goal which this work is built to overcome. If one does a measurement without distinguishing the two channels of the second splitting, the contributions of both sum together, leaving a result which almost cancels itself. The endured signal measured in this case is not the intrinsic strength of the correlation but a residue, because its magnitude and even its sign are set by the balance between the $g \to gg$ and $g \to q\bar{q}$ channels present in the sample. However, this interference is not absent. It is just diluted and this dilution reflects the flavour composition of the selection rather than any property of QCD.

Thus, it is required to isolate a single channel in order to make a reliable measurement of the effect. This work shows the selection procedure of the $g \to q\bar{q}$ splittings, and measures the parameter a_2/a_0 on that subsample alone, replacing the diluted inclusive residue with the coefficient of a single channel. This is precisely what the infrared-safe and collinear-safe flavour labelling of Section 2.2.4 makes possible. IFN tags the flavour, exposed during the declustering, of each splitting, allowing the $g \to q\bar{q}$ channel to be enriched and the $\cos 2\psi$ modulation to be measured as a function of the purity of that enrichment. The channel dependence described here is therefore not just a feature of the physics but the reason the flavour selection developed in this work is required, and it is the procedure that connects the spin observable of the next section to the flavoured-jet algorithm of Section 2.2.4.

2.4.3 The Lund observable $\Delta\psi_{12}$

The effect described in the previous sections can be studied using a concrete observable, given by the Lund declustering of Section 2.3, in order to indentify the two splittings whose planes are to be compared. This construction was proposed in [16], and the observable is the azimuthal angle $\Delta\psi_{12}$ between the hardest primary and secondary branchings of a jet.

The two splittings are selected as it follows. When declustering the primary Lund sequence of a jet, one selects the splitting which passes the Soft Drop condition of equation (2.12) and has the largest relative transverse momentum k_T among all the primary sequence. This splitting is then called the primary splitting. Then, declustering the soft branch of this selected primary splitting, we build the secondary sequence generated by it. Thus, with the same procedure as in the primary sequence, one selects the splitting which passes the Soft Drop condition of equation (2.12) and has the largest relative transverse momentum k_T among all the secondary sequence. This splitting is called the secondary splitting. Each of the two chosen splittings defines a plane in momentum space. The plane $\mathcal{P}_1$ spanned by the branches of the first, and $\mathcal{P}_2$ spanned by the branches of the second. With this, $\Delta\psi_{12}$ is the azimuthal angle between them, illustrated in Figure 2.4.

By construction, $\Delta\psi_{12}$ is the realisation for jets of the abstract angle ψ of equation (2.13). Its distribution therefore inherits the $\cos 2\psi$ modulation derived in Section 2.4.1,

$$\frac{1}{\sigma}\frac{d\sigma}{d\Delta\psi_{12}} \;\propto\; 1 + \frac{a_2}{a_0}\cos 2\Delta\psi_{12}, \tag{2.14}$$

and the coefficient a_2/a_0 extracted from it is the quantity that carries the spin-correlation physics. Two features of the definition are essential to what follows. First, because the second splitting is drawn from the secondary sequence seeded by the first, $\Delta\psi_{12}$ compares precisely the two successive branchings between which the polarisation memory of Section 2.4.1 is transmitted. Second, when that second splitting is the decay of an intermediate gluon, its channel $g \to gg$ or $g \to q\bar{q}$ can be identified by the flavour labelling of Section 2.2.4. Splittings of the form $q \to qg$, in which no intermediate gluon is present,

carry no correlation and act only as a background to be removed. It is this flavour resolution that makes $\Delta\psi_{12}$ not merely a measurement of the inclusive, diluted modulation, but a channel-resolved probe of the effect. By restricting to secondary splittings tagged as $g \to q\bar{q}$, and thereby separating them both from $g \to gg$ splittings and from the correlation-free $q \to qg$ background, the sign cancellation of Section 2.4.2 is undone, and a_2/a_0 can be measured on a single channel.

2.4.4 The three-point energy correlator

Although the lund observable described in the previous section gives a result for the collinear spin correlations of Section 2.4.1, it is not the route to do so. The $\cos 2\psi$ modulation can be accessed through the energy flow within a jet and without any reference to any clustering history, using the three-point energy correlator (EEEC) [17]. Where $\Delta\psi_{12}$ reconstructs the two splitting planes by declustering, the energy correlator reads the effect directly from the angular and energy distribution of the jet's constituents, and the two therefore provide complementary handles on the same underlying physics.

The EEEC measures the energy-weighted distribution of triplets of particles as a function of their relative angles. Concretely, it is defined as the energy-weighted sum over all triplets of particles in the jet,

$$\frac{1}{\sigma}\frac{d^3\Sigma}{d\psi\, d\theta_S\, d\theta_L} = \sum_{i,j,k} \frac{p_{\mathrm{T},i}\, p_{\mathrm{T},j}\, p_{\mathrm{T},k}}{p_{\mathrm{T,jet}}^3}\, \delta(\psi - \psi_{ijk})\, \delta(\theta_S - \theta_{S,ijk})\, \delta(\theta_L - \theta_{L,ijk}), \tag{2.15}$$

where the sum runs over all triplets (i, j, k) of the jet's constituents, each weighted by the product of their transverse momenta normalised to that of the jet. Here θ_S and θ_L are the smallest and largest opening angles of the triplet, and ψ is the azimuthal angle between the plane spanned by the two closest particles and that spanned by the third. Its power for the present purpose is that this last variable retains sensitivity to the azimuthal orientation discussed in Section 2.4.1. This means that the correlator's dependence on ψ carries the same $\cos 2\psi$ modulation, governed by the same interference between gluon polarisation states. Being built from energy weights and angles alone, the correlator is infrared and collinear safe by construction and requires no declustering, no grooming and no jet-flavour assigment.

This makes the energy correlator an independent cross-check on the physics extracted through $\Delta\psi_{12}$. It reaches the same $\cos 2\psi$ effect by a route that shares none of the reconstruction choices on which the Lund observable depends, so that agreement between the two would establish that the measured modulation is a property of the radiation rather than an artefact of any one analysis chain. Note that the coefficient a_2/a_0 extracted from the energy correlator is not expected to coincide numerically with that from $\Delta\psi_{12}$, since the two observables define the azimuthal angle through different objects and carry different sensitivities to the kinematic selection. What the two share is the sign and the qualitative behaviour dictated by the underlying gluon-polarisation interference, and it is at that level that they cross-check one another. The correlator is developed in this work as a complementary probe alongside the primary $\Delta\psi_{12}$ measurement, and its implementation and results are presented in Chapter 4.

3 DATA AND ANALYSIS

The preceding chapter established the observables through which collinear spin correlations can be accessed. Following that, this chapter describes how they are measured in practice and presents the data on which this analysis is built, the software environment in which it runs, and the sequence of steps that leads from the raw available events to the azimuthal distributions of Chapter 4. The goal of this chapter is to reveal all the steps necessary to make this analysis, securing reproducibility. That said, enough detail is given, and the code used is available publicly in [18], so that the analysis can be repeated in full from public inputs alone.

The chapter is organised as follows. Section 3.1 introduces the CERN Open Data initiative and the principles that make an analysis of this kind possible outside the experimental collaborations. Section 3.2 describes the relevant features of the CMS detector, and Section 3.3 the data format and the specific dataset used. Section 3.4 presents the analysis pipeline, from the extraction of jet constituents to the computation of the spin observables. Section 3.5 describes the flavour selection that isolates the $g \to q\bar{q}$ channel, and Section **??** assesses the systematic uncertainties affecting the measurement.

3.1 CERN Open Data

The data analysed in this work were obtained through the CERN Open Data Portal [2], the platform through which the LHC experiments release reconstructed collision data and the associated simulation to the public. Launched in 2014, the portal was the first initiative of its kind in high-energy physics, and it has since grown into a major repository holding data from the ALICE, ATLAS, CMS and LHCb collaborations. The first releases came from the CMS experiment, and it is CMS data that this work uses.

The CERN Open Data Portal does not provide detector output but data at the same level of reconstruction used in collaborations' own published analysis. These datasets include physics objects such as jets, tracks and reconstructed particles, with the calibration already applied and run quality information required to use them correctly. Alongside the collision data, the portal also releases the corresponding Monte Carlo simulations, the software environment necessary to analyse the datasets and documentation describing their format and content. All this features of the portal make it possible for any user to use their data for their own analysis and interpret the results.

The importance of this portal in this work is twofold. First, as made clear in this chapter's introduction, it makes the measurement reproducible. Every input from the collision and respective simulated datasets are publicly available, so that the analysis can be repeated in full by anyone. Second, the portal specifies, for each dataset, the steps required to use them correctly. For each dataset used in this work, CMS provides a list of available and validated runs, with the proper calibrations and standard energy corrections and residual effects, applied by them.

This type of analysis are feasible outside the collaborations has been demonstrated by a growing body of work built on CMS Open Data. Studies using released CMS samples have addressed problems ranging from the discrimination of quark and gluon initiated jets [19] to the measurement of N-point energy correlators inside jets [20], the latter closely related to the observables of the present work. These analyses establish both that public data retain the information needed for substructure measurements

and that results of genuine physics interest can be extracted from them. The present work is situated in this line. It applies the same public data to the measurement of collinear spin correlations, an effect not previously accessed in this way.

The only small downside of working with public data is the choice of software environment. The released CMS data are stored in formats tied to the experiment's reconstruction software, and reading them requires a compatible environment. CMS provides this in the form of a dedicated container image, inside docker [21], that reproduces the software stack in which the data were generated. In this work that environment was used to read the released files and extract the jet and constituent information required by the analysis, which was then carried out in a lighter, independent environment. The details of this arrangement are given in Section 3.3 and Section 3.4.

3.2 The CMS detector

The collision data used and analysed in this work was recorded by the Compact Muon Solenoid (CMS). This giant detector, alongside with ATLAS, is a general-purpose detector, at the LHC [22]. It is a cylindrical apparatus with 21.6 m in length and 14.6 m of diameter, built in concentric layers around the beam pipe with the characteristic feature of having a superconducting solenoid, with 6 m in internal diameter, capable of producing a uniform 3.8 T magnetic field, that bends the trajectories of charged particles allowing for their momenta to be measured. Its layers are capable of capturing relevant features of this analysis and they will be described below in the order of a particle transversing them.

The CMS coordinate system has the origin centered at the nominal collision point inside the detector, its y-axis points vertically upwards and the x-axis points radially inwards towards the center of the LHC. This makes the z-axis point along the beam axis towards Jura mountains from LHC Point 5 [22]. The polar angle θ is measured from the z axis, The azimuthal angle ϕ is measured from the x-axis in the x-y plane and the radial coordinate in this plane is denoted by r. The polar angle θ is measured from the z axis, although it is conventional to use the pseudorapidity $\eta = -\ln\tan(\theta/2)$, because differences in η are invariant under boosts along the beam. The transverse momentum is denoted as p_T and it is the component of momentum in the plane perpendicular to the beam. These are the variables which jets and their constituents are used on this work.

Starting from the layer closest to the interaction point, the inner-tracker is an all-silicon system of pixel and strip detectos covering $|\eta| < 2.5$. This layer is placed inside the solenoid's field and it reconstructs the trajectories of charged particles, measures their momenta and detects the primary interaction vertex. Since substructure observables used in this work are built from the trajectories and momenta of a jet's charged constituents, this tracker is the subdetector which these measurements directly rely on.

There are two calorimeters surrounding the inner-tracker. The electromagnetic calorimeter (ECAL) uses lead tungstate scintillating crystal to measure the energy of electrons and photons. Surrounding it, the hardron calorimeter (HCAL), a brass/scintillator device, measures the energy of the hadrons. Together, these two calorimeters cover the central region and, with more calorimeters forward extending the coverage to $|\eta| < 5$, give CMS a near complete solid-angle coverage needed to measure the flow of energy in an event. The muon system outside the solenoid, plays no important role for this analysis.

When CMS reconstructs jets, they don't rely on just one of these subdetectors isolated. Instead, they use their own particle-flow algorithm, tuned specifically for the CMS detector [23], which combines signals from all the subdetectors, including tracks from the inner-tracker and energy deposits from the

calorimeters, to to reconstruct a list of individual particles in each event, the particle-flow candidates. It is these candidates that are the inputs to the jets' clustering algorithms of Section 2.2.1, and their charged component measured by the tracker that most directly determines the directions from which the substructure observables are built. This particle-flow algorithm is therefore the point at which the detector described here connects to the analysis that follows, and the data format in which its output is released is the subject of the next section. The following figure 3.1, gives a solid perspective of the CMS detector.

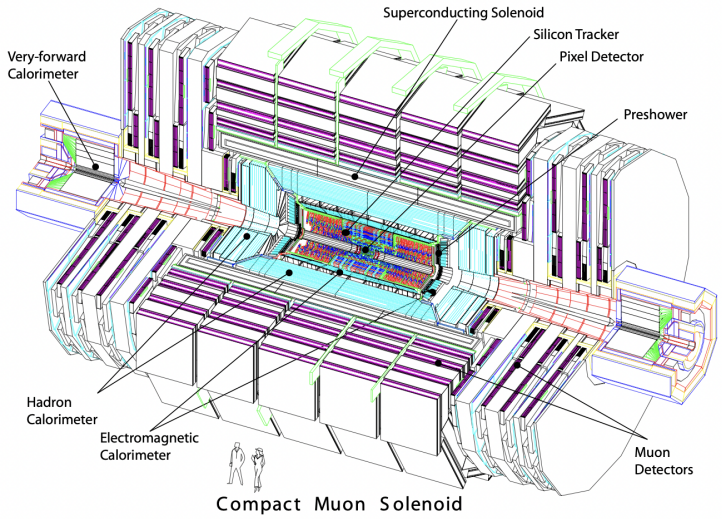


Figure 3.1 – A perspective view of the CMS detector [22].

3.3 Data format and datasets

The CMS released reconstructed data in the CERN Open data format comes in a hierarchy of formats. Each format sacrifices completeness for size making the choice of format the first decision required for an analysis using their datasets, since it fixes what information is available and what variables are at hand. The three formats relevant here are summarised in Table 3.1.

Table 3.1 – The three CMS data-tier formats, in decreasing order of size and detail. Sizes are approximate and quoted relative to the baseline AOD.

Format	Content	Size
AOD	High-level physics objects plus substantial lower-level reconstruction detail.	baseline
MiniAOD	Analysis-level physics objects similar to AOD but slimmed: many collections are reduced, extra information is packed into lighter objects, and less commonly used low-level detail is removed.	$\sim 1/10$ of AOD
NanoAOD	Compact ntuple-style content: a limited set of per-object observables and precomputed IDs; little to no low-level detector detail (e.g., Particle Flow candidates dropped).	$\sim 1/20$ of MiniAOD

For this work, the requirement is having access to the constituents of a jet. The substructure observables of Chapter 2 are built by declustering a jet into its constituents, so this analysis requires the individual particle-flow candidates that build the jet and not just a summary of the jet's kinematics. This immediality invalidates the NanoAOD data format since the particle-flow candidates are not stored there. The AOD format stores the particle-flow candidates, however it also stores other low-level reconstruction details

which this analysis does not need making it occupy a large volume of space. This leaves the MiniAOD data format, which occupies the middle ground of space and still stores particle-flow candidates required for substructure while discarding the low-level information that is not needed. Therefore, it is for this reason that the MiniAOD data format was the choice made for the format to be used throughout this work.

Between all the MiniAOD datasets, this work analysed the JetHT primary dataset from the 2016 data-taking period, in MiniAOD format [24]. Its characteristics are collected in Table 3.2. The data are proton–proton collisions at a centre-of-mass energy of $\sqrt{s} = 13$ TeV, recorded during run period G of 2016 and released as part of the Ultra-Legacy reprocessing, the final and most complete calibration of the Run 2 data.

Table 3.2 – Characteristics of the JetHT primary dataset [24].

Property	Value
Dataset	`/JetHT/Run2016G-UL2016_MiniAODv2-v2/MINIAOD`
Collision	proton–proton, $\sqrt{s} = 13$ TeV
Run period	2016G (runs 278820–280385)
Events	1.21×10^8
Size	4.1 TiB (1236 files)

The JetHT dataset is well suited for this work analysis since it is enriched in jet events. CMS records events online that pass a two-level trigger system, composed by a hardware Level-1 trigger followed by a software High Level Trigger (HLT) [25]. The events of this dataset are then those accepted by a set of HLT paths, which require either a large scalar sum of jet transverse momenta or the presence of one or two energetic jets. As a result the sample is dominated by the high-transverse-momentum, high-multiplicity final states in which jet substructure is most informative. Since the dataset is served by triggers based on large-radius jets, events are required to pass one of the `HLT_AK8PFJet` paths. The analysis is restricted to jets with transverse momentum above 700 GeV, where the unprescaled paths of highest threshold, `HLT_AK8PFJet450` and `HLT_AK8PFJet500`, are almost fully efficient. In this region any jet passing the selection also satisfies these paths, so the trigger requirement introduces no turn-on bias, and the sample sits in the hard, perturbative regime motivated in Section 2.1.

Furthermore, as required for all analysis of this dataset, the data is restricted to a list of runs certified as good physics, provided with the dataset release. So, events not in these certified runs do not enter this analysis. In addition, jets in this released sample do not require any energy calibrations to be applied to them, since they already carry the standard jet energy corrections applied by CMS in the Ultra-Legacy reprocessing.

Alongside the JetHT collision dataset, this analysis uses a simulated sample of QCD multijet events. This is because the flavour of a splitting inside a jet cannot be accessed in data, so it is not possible to determine the secondary splitting channel. What the simulation provides is the set of generated partons, carrying flavour, which are clustered in the same way as the reconstructed particles in data. First the partons are clustered into jets with the anti-k_T algorithm of section 2.2.2 and then, through the Cambridge/Aachen algorithm of Section 2.2.3 and the IFN algorithm of Section 2.2.4, into a branching history in which each splitting acquires a flavour. It is from this process that the channel of each splitting, $g \to q\bar{q}$, $g \to gg$ or $q \to qg$, is labelled. This flavour labelling process and how it is further used in collision data is discussed in Section 3.5. The sample used is a QCD multijet dataset generated with Pythia 8 [**Sjostrand:2014zea**] using the CP5 tune [26], passed through the full simulation of the CMS detector and released in the same format as the collision data (MiniAODSIM) [27].

3.4 Analysis pipeline

3.5 Flavour selection

3.6 Systematic uncertainties

4 RESULTS AND DISCUSSION

5 CONCLUSIONS

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